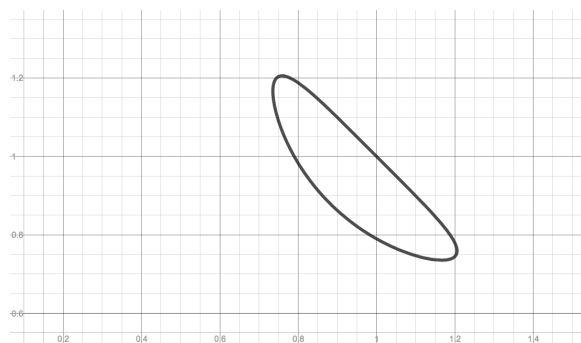


Problem 3

Problem 3

Problem 1 A part of the curve with equation $\cos(\pi xy) + x + y = 1$ is sketched below.



- (a) Use the implicit differentiation to find the derivative dy/dx .

Explanation. Implicitly differentiate equation:

$$\frac{d}{dx}(\cos(\pi xy) + x + y) = \frac{d}{dx}(1) \implies -\sin(\pi xy) \cdot (\pi y + \pi x \frac{dy}{dx}) + 1 + \frac{dy}{dx} = 0$$

Solve for $\frac{dy}{dx}$:

$$\begin{aligned} -\sin(\pi xy) \cdot (\pi y + \pi x \frac{dy}{dx}) + 1 + \frac{dy}{dx} &= 0 \implies -\pi y \sin(\pi xy) - \pi x \frac{dy}{dx} \sin(\pi xy) + 1 + \frac{dy}{dx} = 0 \\ \implies (1 - \pi x \sin(\pi xy)) \frac{dy}{dx} &= \pi y \sin(\pi xy) - 1 \\ \implies \frac{dy}{dx} &= \frac{\pi y \sin(\pi xy) - 1}{1 - \pi x \sin(\pi xy)} \quad \text{if } 1 - \pi x \sin(\pi xy) \neq 0 \end{aligned}$$

- (b) Consider the point $(1, 1)$. Show (algebraically) that this point lies on the curve.

Explanation. $(1, 1)$ lies on this curve:

$$\begin{aligned} \cos(\pi \cdot 1 \cdot 1) + 1 + 1 &= \cos(\pi) + 2 \\ &= -1 + 2 = 1 \end{aligned}$$

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- (c) Find the equation of the line tangent to the curve at $(1, 1)$. Draw this line in the figure above.

Explanation. Slope of tangent line:

$$\begin{aligned} \left[\frac{dy}{dx} \right]_{(1,1)} &= \frac{\pi \cdot (1) \cdot \sin(\pi(1) \cdot 1) - 1}{1 - \pi \cdot (1) \cdot \sin(\pi(1) \cdot (1))} \\ &= \frac{\pi \sin(\pi) - 1}{1 - \pi \sin(\pi)} \\ &= \frac{-1}{1} = -1 \end{aligned}$$

Equation of tangent line:

$$y - 1 = -1(x - 1) \implies y = -x + 2$$

Graph of tangent line:

